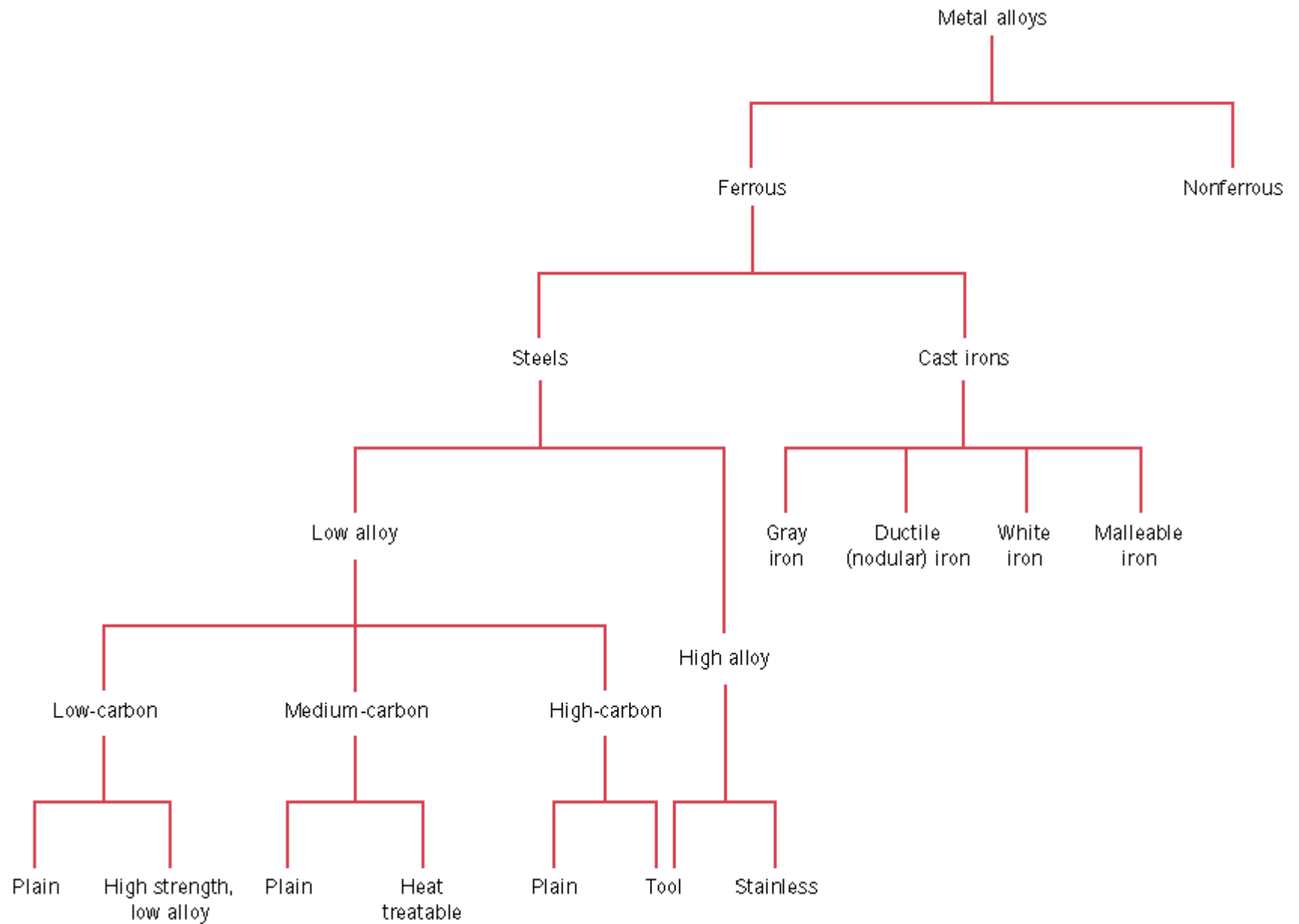


Applications of Ferrous and Non-Ferrous alloys



Nomenclature of Steels

- Historically, many methods for identifying alloys by their composition have been developed
- The commonly used schemes are those developed by AISI/SAE and ASTM
 - The American Iron and Steel Institute (AISI) and the Society of Automotive Engineers (SAE)
 - American Society for Testing and Materials (ASTM)
- European countries, Japan, Russia etc. developed their own schemes
- In order to avoid confusion, the Universal/Unified Numbering System (UNS) was developed

AISI/SAE Classification of Steels

- A four digit description
 - First two digits identify the alloy type
 - Last two digits indicate the carbon content
 - For example
 - AISI/SAE 1020 steel is a plain carbon steel (10xx) which has 0.20 wt.% carbon (xx20)
- Plain carbon steel (10xx) are inexpensive, but have several limitations including:
 - Poor hardenability because the critical cooling rate is very high
 - Rapid cooling leads to distortion and cracking
 - Poor corrosion resistance
 - Poor impact resistance at low temperature
- Alloy steels were developed to address these issues
 - Alloying changes the eutectoid composition, the eutectoid carbon content and the critical cooling rate
 - These alloys are more expensive, but a better combination of properties is obtained

AISI/SAE Classification of Steels

AISI/SAE Designation ^a	UNS Designation	Composition Ranges (wt% of Alloying Elements in Addition to C) ^b			
		Ni	Cr	Mo	Other
10xx, Plain carbon	G10xx0				
11xx, Free machining	G11xx0				0.08–0.33S
12xx, Free machining	G12xx0				0.10–0.35S, 0.04–0.12P
13xx	G13xx0				1.60–1.90Mn
40xx	G40xx0			0.20–0.30	
41xx	G41xx0		0.80–1.10	0.15–0.25	
43xx	G43xx0	1.65–2.00	0.40–0.90	0.20–0.30	
46xx	G46xx0	0.70–2.00		0.15–0.30	
48xx	G48xx0	3.25–3.75		0.20–0.30	
51xx	G51xx0		0.70–1.10		
61xx	G61xx0		0.50–1.10		0.10–0.15V
86xx	G86xx0	0.40–0.70	0.40–0.60	0.15–0.25	
92xx	G92xx0				1.80–2.20Si

^aThe carbon concentration, in weight percent times 100, is inserted in the place of “xx” for each specific steel.

^bExcept for 13xx alloys, manganese concentration is less than 1.00 wt%.

Except for 12xx alloys, phosphorus concentration is less than 0.35 wt%.

Except for 11xx and 12xx alloys, sulfur concentration is less than 0.04 wt%.

Except for 92xx alloys, silicon concentration varies between 0.15 and 0.35 wt%.

UNS uses the AISI/SAE designation with a letter before and a “0” after the 4 digits
The letter identifies the alloy group

Overview of UNS

- **Axxxxx** - Aluminum Alloys
- **Cxxxxx** - Copper Alloys, including Brass and Bronze
- **Fxxxxx** - Iron, including Ductile Irons and Cast Irons
- **Gxxxxx** - Carbon and Alloy Steels
- **Hxxxxx** - Steels - AISI H Steels
- **Jxxxxx** - Steels - Cast
- **Kxxxxx** - Steels, including Maraging, Stainless, HSLA, Iron-Base Superalloys
- **L5xxxx** - Lead Alloys, including Babbit Alloys and Solders
- **M1xxxx** - Magnesium Alloys
- **Nxxxxx** - Nickel Alloys
- **Rxxxxx** - Refractory Alloys
 - **R03xxx**- Molybdenum Alloys
 - **R04xxx**- Niobium (Columbium) Alloys
 - **R05xxx**- Tantalum Alloys
 - **R3xxxx**- Cobalt Alloys
 - **R5xxxx**- Titanium Alloys
 - **R6xxxx**- Zirconium Alloys
- **Sxxxxx** - Stainless Steels, including Precipitation Hardening and Iron-Based Superalloys
- **Txxxxx** - Tool Steels
- **Zxxxxx** - Zinc Alloys

AISI/SAE, ASTM, UNS

- ASTM developed a parallel classification, starting with a letter A followed by numbers and other descriptors

<i>Designation^a</i>		<i>Composition (wt%)^b</i>		
<i>AISI/SAE or ASTM Number</i>	<i>UNS Number</i>	<i>C</i>	<i>Mn</i>	<i>Other</i>
<i>Plain Low-Carbon Steels</i>				
1010	G10100	0.10	0.45	
1020	G10200	0.20	0.45	
A36	K02600	0.29	1.00	0.20 Cu (min)
A516 Grade 70	K02700	0.31	1.00	0.25 Si
<i>High-Strength, Low-Alloy Steels</i>				
A440	K12810	0.28	1.35	0.30 Si (max), 0.20 Cu (min)
A633 Grade E	K12002	0.22	1.35	0.30 Si, 0.08 V, 0.02 N, 0.03 Nb
A656 Grade 1	K11804	0.18	1.60	0.60 Si, 0.1 V, 0.20 Al, 0.015 N

Mechanical Characteristics of Hot-Rolled Material and Typical Applications for Various Plain Low-Carbon and High-Strength, Low-Alloy Steels

<i>AISI/SAE or ASTM Number</i>	<i>Tensile Strength [MPa (ksi)]</i>	<i>Yield Strength [MPa (ksi)]</i>	<i>Ductility [%EL in 50 mm (2 in.)]</i>	<i>Typical Applications</i>
<i>Plain Low-Carbon Steels</i>				
1010	325 (47)	180 (26)	28	Automobile panels, nails, and wire
1020	380 (55)	205 (30)	25	Pipe; structural and sheet steel
A36	400 (58)	220 (32)	23	Structural (bridges and buildings)
A516 Grade 70	485 (70)	260 (38)	21	Low-temperature pres- sure vessels
<i>High-Strength, Low-Alloy Steels</i>				
A440	435 (63)	290 (42)	21	Structures that are bolted or riveted
A633 Grade E	520 (75)	380 (55)	23	Structures used at low ambient temperatures
A656 Grade 1	655 (95)	552 (80)	15	Truck frames and rail- way cars

Typical Applications and Mechanical Property Ranges for Oil-Quenched and Tempered Plain Carbon and Alloy Steels

<i>AISI Number</i>	<i>UNS Number</i>	<i>Tensile Strength [MPa (ksi)]</i>	<i>Yield Strength [MPa (ksi)]</i>	<i>Ductility [%EL in 50 mm (2 in.)]</i>	<i>Typical Applications</i>
<i>Plain Low-Carbon Steels</i>					
1040	G10400	605–780 (88–113)	430–585 (62–85)	33–19	Crankshafts, bolts
1080 ^a	G10800	800–1310 (116–190)	480–980 (70–142)	24–13	Chisels, hammers
1095 ^a	G10950	760–1280 (110–186)	510–830 (74–120)	26–10	Knives, hacksaw blades
<i>Alloy Steels</i>					
4063	G40630	786–2380 (114–345)	710–1770 (103–257)	24–4	Springs, hand tools
4340	G43400	980–1960 (142–284)	895–1570 (130–228)	21–11	Bushings, aircraft tubing
6150	G61500	815–2170 (118–315)	745–1860 (108–270)	22–7	Shafts, pistons, gears

^a Classified as high-carbon steels.

Tool Steels

<i>AISI Number</i>	<i>UNS Number</i>	<i>Composition (wt%)^a</i>						<i>Typical Applications</i>
		<i>C</i>	<i>Cr</i>	<i>Ni</i>	<i>Mo</i>	<i>W</i>	<i>V</i>	
M1	T11301	0.85	3.75	0.30 max	8.70	1.75	1.20	Drills, saws; lathe and planer tools
A2	T30102	1.00	5.15	0.30 max	1.15	—	0.35	Punches, embossing dies
D2	T30402	1.50	12	0.30 max	0.95	—	1.10 max	Cutlery, drawing dies
O1	T31501	0.95	0.50	0.30 max	—	0.50	0.30 max	Shear blades, cutting tools
S1	T41901	0.50	1.40	0.30 max	0.50 max	2.25	0.25	Pipe cutters, concrete drills
W1	T72301	1.10	0.15 max	0.20 max	0.10 max	0.15 max	0.10 max	Blacksmith tools, wood-working tools

AISI designation has a letter and a number.

The letter describes the application

– M (high speed machine tool), H (hot working)

The letter describes the heat treatment

– A (air hardening), O (oil quenching), W (water quenching)

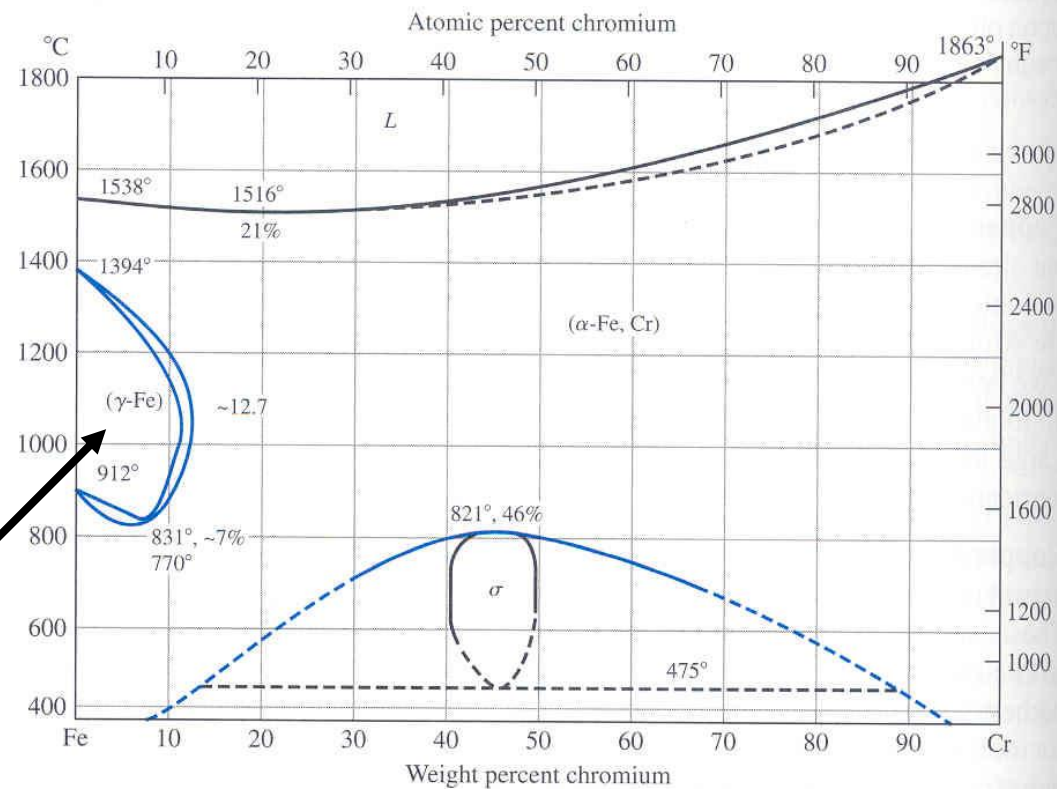
UNS designation – all tool steels start with a “T”

Stainless Steels

- Excellent corrosion resistance
- Contain 12 to 30% Chromium
 - Cr oxidizes easily and forms a thin continuous layer of oxide that prevents further oxidation of the metal
- Cr is a ferrite stabilizer

- ***Ferritic Stainless Steels*** are essentially Fe-Cr Alloys
- Ferrite phase (bcc structure)
- Inexpensive, high strength

Austenite is restricted to a small region of the phase diagram



Stainless Steels

- Austenitic Stainless Steels
 - Nickel is an austenite stabilizer. The addition of both Cr and Ni results in the austenite (γ , fcc) phase being retained to room temperature
 - The austenite phase is very formable (fcc structure)
 - Ni makes these alloys expensive
- Martensitic Stainless Steels
 - Have both Cr and C
 - There is more Cr than in ferritic SS since Cr tends to form Cr_{23}C_6 , which removes available Cr for corrosion protection
 - Can be heat treated to high strength

AISI Number	UNS Number	Composition (wt%) ^a	Condition ^b	Mechanical Properties			Typical Applications
				Tensile Strength [MPa (ksi)]	Yield Strength [MPa (ksi)]	Ductility [%EL in 50 mm (2 in.)]	
Ferritic							
409	S40900	0.08 C, 11.0 Cr, 1.0 Mn, 0.50 Ni, 0.75 Ti	Annealed	380 (55)	205 (30)	20	Automotive ex- haust compo- nents, tanks for agricul- tural sprays
446	S44600	0.20 C, 25 Cr, 1.5 Mn	Annealed	515 (75)	275 (40)	20	Valves (high tempera- ture), glass molds, com- bustion chambers
Austenitic							
304	S30400	0.08 C, 19 Cr, 9 Ni, 2.0 Mn	Annealed	515 (75)	205 (30)	40	Chemical and food pro- cessing equipment, cryogenic vessels
316L	S31603	0.03 C, 17 Cr, 12 Ni, 2.5 Mo, 2.0 Mn	Annealed	485 (70)	170 (25)	40	Welding con- struction

UNS letter S indicates stainless steel

AISI Number	UNS Number	Composition (wt%) ^a	Condition ^b	Mechanical Properties			Typical Applications
				Tensile Strength [MPa (ksi)]	Yield Strength [MPa (ksi)]	Ductility [%EL in 50 mm (2 in.)]	
Martensitic							
410	S41000	0.15 C, 12.5 Cr, 1.0 Mn	Annealed Q & T	485 (70)	275 (40)	20	Rifle barrels, cutlery, jet engine parts
				825 (120)	620 (90)	12	
440A	S44002	0.70 C, 17 Cr, 0.75 Mo, 1.0 Mn	Annealed Q & T	725 (105)	415 (60)	20	Cutlery, bear- ings, surgical tools
				1790 (260)	1650 (240)	5	
Precipitation Hardenable							
17-7PH	S17700	0.09 C, 17 Cr, 7 Ni, 1.0 Al, 1.0 Mn	Precipita- tion hardened	1450 (210)	1310 (190)	1–6	Springs, knives, pres- sure vessels

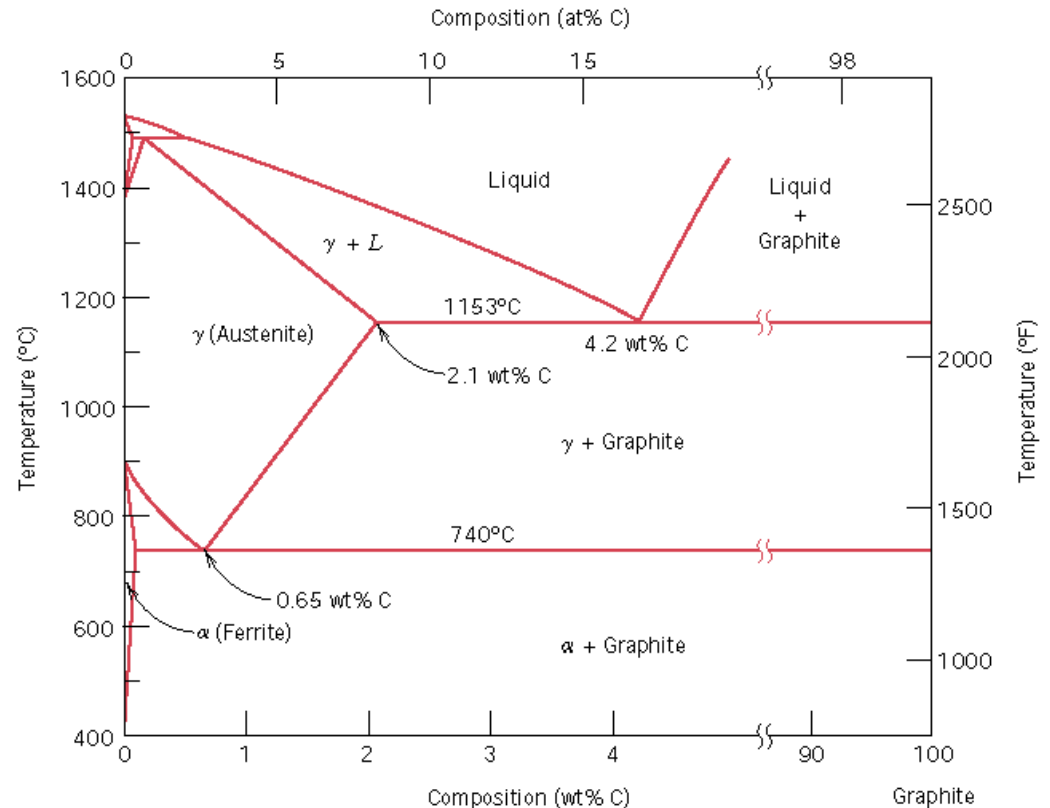
^a The balance of the composition is iron.

^b Q & T denotes quenched and tempered.

Source: Adapted from *ASM Handbook*, Vol. 1, *Properties and Selection: Irons, Steels, and High-Performance Alloys*, 1990. Reprinted by permission of ASM International, Materials Park, OH.

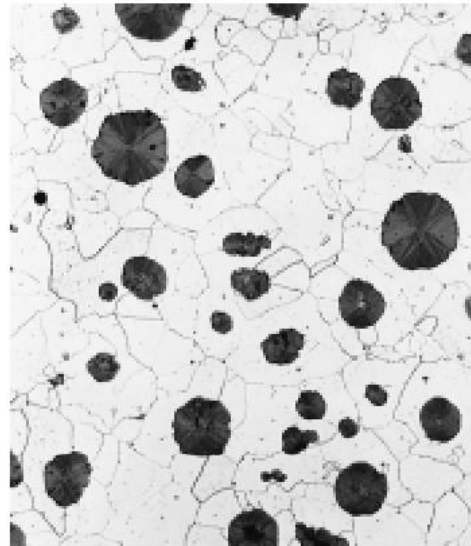
Cast Iron

- Fe-C alloys with 2-4% C
- 1-3% Si is added to improve castability
- Phase diagram shows graphite rather than Fe_3C since C may be present in the form of both graphite and cementite
- Temperatures and compositions are different from the Fe- Fe_3C diagram
- Features:
 - Low melting temperature (1153°C to 1400°C)
 - Low shrinkage
 - Easily machinable
 - Low impact resistance
 - Low ductility



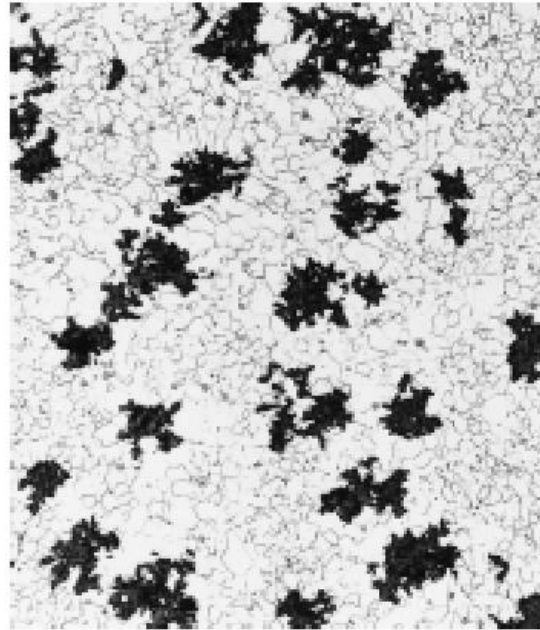
Cast Irons

- Types
 - Gray cast iron
 - Carbon in the form of graphite flakes
 - 2.5 – 4% C and 1 – 3% Si (Promotes formation of graphite)
 - Nodular cast iron
 - Carbon in the form of spherical graphite nodules
 - 3-4% C and 1.8 – 2.8 % Si + Mg or Ce, and low impurities

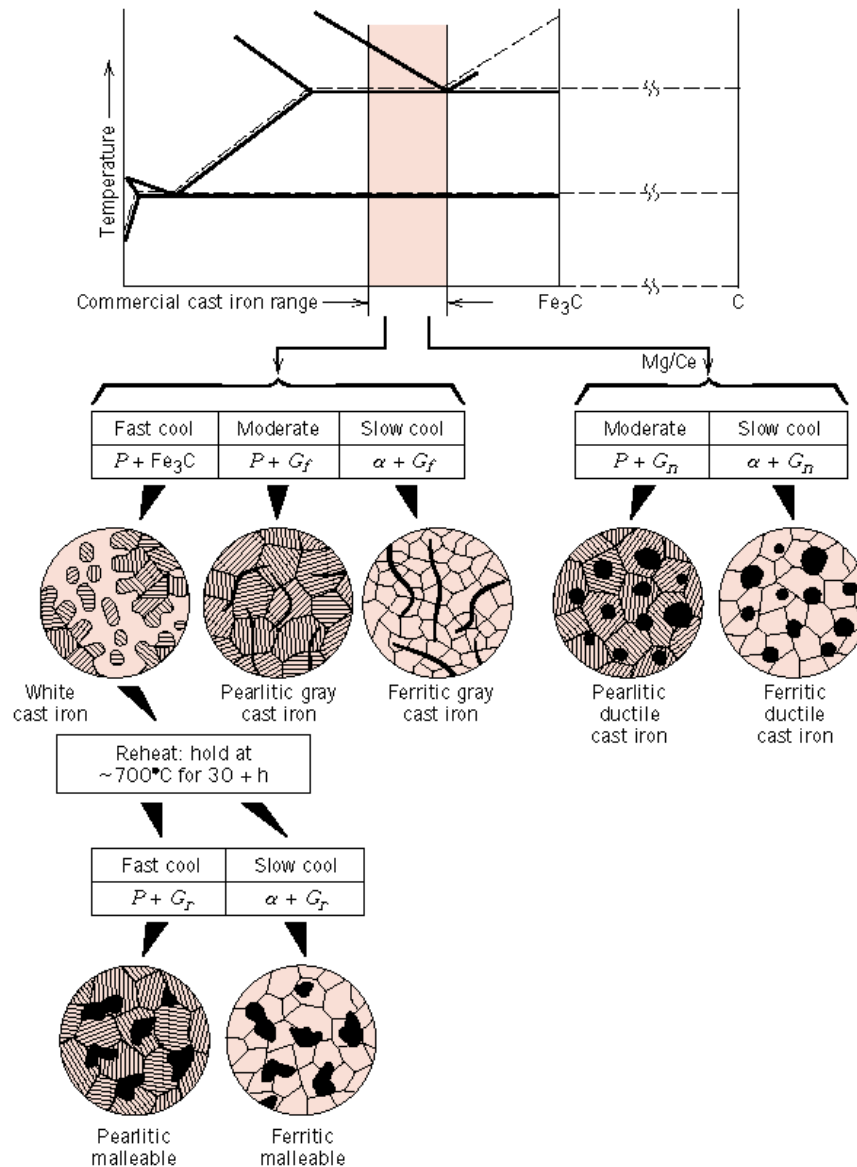


Cast Irons

- Types
 - White cast iron
 - Carbon in the form of cementite
 - Malleable cast iron
 - Carbon in the form of irregular graphite nodules
 - Obtained by heat treating white cast iron



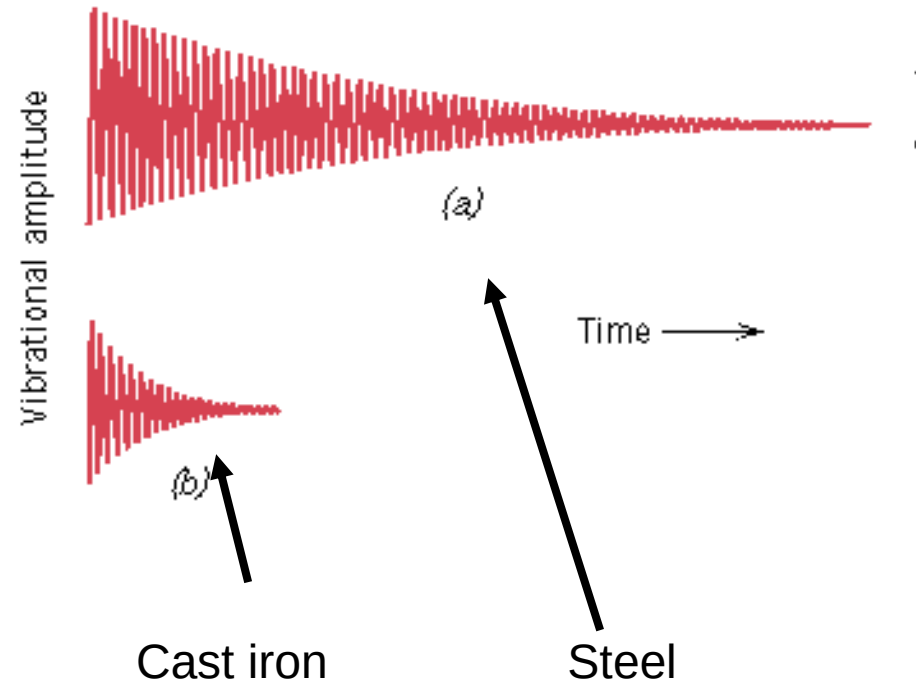
Cast Irons



- The microstructure of the iron rich matrix can be modified by heat treatment
 - Pearlite
 - Ferrite
- Gray cast iron
 - Fracture surface appears gray because of graphite flakes
- White cast iron
 - Fracture surface appears white (shiny)

Cast Irons

- White cast iron has no other use than to be starting material for malleable cast iron
- In the other forms of cast iron, carbon is in the form of graphite
 - The graphite flakes absorb vibration
 - Lubricate during machining
 - Fracture initiation sites



Grade	UNS Number	Composition (wt%)*	Matrix Structure	Mechanical Properties			Typical Applications
				Tensile Strength [MPa (ksi)]	Yield Strength [MPa (ksi)]	Ductility [%EL in 50 mm (2 in.)]	
Gray Iron							
SAE G1800	F10004	3.40–3.7 C, 2.55 Si, 0.7 Mn	Ferrite + Pearlite	124 (18)	—	—	Miscellaneous soft iron castings in which strength is not a primary consideration
SAE G2500	F10005	3.2–3.5 C, 2.20 Si, 0.8 Mn	Ferrite + Pearlite	173 (25)	—	—	Small cylinder blocks, cylinder heads, pistons, clutch plates, transmission cases
SAE G4000	F10008	3.0–3.3 C, 2.0 Si, 0.8 Mn	Pearlite	276 (40)	—	—	Diesel engine castings, liners, cylinders, and pistons
Ductile (Nodular) Iron							
ASTM A536 60-40-18	F32800	3.5–3.8 C, 2.0–2.8 Si, 0.05 Mg, <0.20 Ni, <0.10 Mo	Ferrite	414 (60)	276 (40)	18	Pressure-containing parts such as valve and pump bodies
100-70-03	F34800		Pearlite	689 (100)	483 (70)	3	High-strength gears and machine components
120-90-02	F36200		Tempered martensite	827 (120)	621 (90)	2	Pinions, gears, rollers, slides
Malleable Iron							
32510	F22200	2.3–2.7 C, 1.0–1.75 Si, <0.55 Mn	Ferrite	345 (50)	224 (32)	10	General engineering service at normal and elevated temperatures
45006	—	2.4–2.7 C, 1.25–1.55 Si, <0.55 Mn	Ferrite + Pearlite	448 (65)	310 (45)	6	

ASTM – specification by strength and ductility
 UNS – Letter F indicates cast iron

Copper Alloys

- General properties of Copper:
 - Good electrical and thermal conduction
 - ease of fabrication
 - corrosion resistance
 - medium strength
- UNS Classification
 - C followed by 5 digits
 - Numbers C10100 to C79900 designate wrought alloys
 - Numbers C80000 to C99900 designate casting alloys
- Electrolytic tough pitch copper (C11000) is the least expensive and used in production of wire, rod, and strip.
 - Has 0.04% oxygen
 - $\text{Cu}_2\text{O} + \text{H}_2 \longrightarrow 2\text{Cu} + \text{H}_2\text{O}$ at 400°C causing blisters
- Copper cast in controlled reducing atmosphere to form OFHC copper (C10200)

UNS Classification of Copper Alloys

Wrought alloys

C1xxxx	Coppers [†] and high-copper alloys [†]
C2xxxx	Copper-zinc alloys (brasses)
C3xxxx	Copper-zinc-lead alloys (leaded brasses)
C4xxxx	Copper-zinc-tin alloys (tin brasses)
C5xxxx	Copper-tin alloys (phosphor bronzes)
C6xxxx	Copper-aluminum alloys (aluminum bronzes), copper-silicon alloys (silicon bronzes) and miscellaneous copper-zinc alloys
C7xxxx	Copper-nickel and copper-nickel-zinc alloys (nickel silvers)

Cast alloys

C8xxxx	Cast coppers, cast high-copper alloys, cast brasses of various types, cast manganese-bronze alloys, and cast copper-zinc-silicon alloys
C9xxxx	Cast copper-tin alloys, copper-tin-lead alloys, copper-tin-nickel alloys, copper-aluminum-iron alloys, and copper-nickel-iron and copper-nickel-zinc alloys.

Copper Alloys

- Cu-Zn Brass
 - Cu-Zn form substitutional solid solutions up to 35% Zn.
 - Cartridge brass (70Cu 30Zn) is single phase
 - Muntz brass (60Cu 40Zn) is two phase.
 - Zinc (0.5 to 3%) is always added to copper to increase machinability
- Cu-Sn Bronzes
 - 1 to 10% tin with Cu to form solid solution strengthened alloys.
 - Stronger and less corrosive than Cu-Zn bronzes.
 - Up to 16% Sn is added to alloys that are used for high strength bearings.
- Cu-Be alloys
 - 0.6 to 2% Be and 0.2 – 2.5 % Cobalt with copper.
 - Can be heat treated and cold worked to produce very strong (1463 MPa) bronzes.
 - Excellent corrosion resistance and fatigue properties.
 - Used in springs, diaphragms, valves etc.

Alloy Name	UNS Number	Composition (wt%)*	Condition	Mechanical Properties			Typical Applications
				Tensile Strength [MPa (ksi)]	Yield Strength [MPa (ksi)]	Ductility [%EL in 50 mm (2 in.)]	
Wrought Alloys							
Electrolytic tough pitch	C11000	0.04 O	Annealed	220 (32)	69 (10)	45	Electrical wire, rivets, screening, gaskets, pans, nails, roofing
Beryllium copper	C17200	1.9 Be, 0.20 Co	Precipitation hardened	1140–1310 (165–190)	690–860 (100–125)	4–10	Springs, bellows, firing pins, bushings, valves, diaphragms
Cartridge brass	C26000	30 Zn	Annealed	300 (44)	75 (11)	68	Automotive radiator cores, ammunition components, lamp fixtures, flashlight shells, kickplates
			Cold-worked (H04 hard)	525 (76)	435 (63)	8	
Phosphor bronze, 5% A	C51000	5 Sn, 0.2 P	Annealed	325 (47)	130 (19)	64	Bellows, clutch disks, diaphragms, fuse clips, springs, welding rods
			Cold-worked (H04 hard)	560 (81)	515 (75)	10	
Copper-nickel, 30%	C71500	30 Ni	Annealed	380 (55)	125 (18)	36	Condenser and heat-exchanger components, saltwater piping
			Cold-worked (H02 hard)	515 (75)	485 (70)	15	
Cast Alloys							
Leaded yellow brass	C85400	29 Zn, 3 Pb, 1 Sn	As cast	234 (34)	83 (12)	35	Furniture hardware, radiator fittings, light fixtures, battery clamps
Tin bronze	C90500	10 Sn, 2 Zn	As cast	310 (45)	152 (22)	25	Bearings, bushings, piston rings, steam fittings, gears
Aluminum bronze	C95400	4 Fe, 11 Al	As cast	586 (85)	241 (35)	18	Bearings, gears, worms, bushings, valve seats and guards, pickling hooks

Aluminum Alloys

- Grouped into Wrought and Cast Alloys
- Wrought Alloys – mechanically worked to final shape
 - 4 digits based on major alloying elements.
 - First digit: major group of alloying elements
 - Second digit: impurity limits
 - Last two digits: identify specific alloy
- Cast Alloys – cast to final shape
 - 4 digits with a period between the third and fourth digit
 - Compositions optimized for casting and mechanical properties
- Alloy designations sometimes preceded with Aℓ or AA
- Also classified into heat-treatable and non-heat treatable alloys
 - Heat treatable alloys are strengthened by precipitation hardening
 - Non-heat treatable alloys are used in the as-cast condition or can be work hardened

Classification of wrought aluminum alloys

Aluminum, 99.00% minimum and greater	1xxx
Aluminum alloys grouped by major alloying elements:	
Copper	2xxx
Manganese	3xxx
Silicon	4xxx
Magnesium	5xxx
Magnesium and silicon	6xxx
Zinc	7xxx
Other element	8xxx
Unused series	9xxx

Non-heat treatable aluminum alloys

- 1xxx alloys : 99% Al + Fe + Si + 0.12% Cu
 - Tensile strength = 90 MPa
 - Used for sheet metals
- 3xxx alloys : Mn principle alloying element
 - AA3003 = AA1100 + 1.25% Mn
 - Tensile strength = 110 MPa
 - General purpose alloy
- 5xxx alloys: Al + up to 5% Mg
 - AA5052 = Al + 2.5%Mg + 0.2% Cr
 - Tensile strength = 193 MPa
 - Used in bus, truck and marine sheet metals.

Heat treatable aluminum alloys

- 2xxx alloys : Al + Cu + Mg
 - AA2024 = Al + 4.5% Cu + 1.5% Mg + 0.6% Mn
 - Strength = 442 MPa
 - Used for aircraft structures.
- 6xxx alloys: Al + Mg + Si
 - AA6061 = Al + 1% Mg + 0.6% Si + 0.3% Cu + 0.2% Cr
 - Strength = 290 MPa
 - Used for general purpose structures.
- 7xxx alloys: Al + Zn + Mg + Cu
 - AA7075 = Al + 5.6% Zn + 2.5% Mg + 1.6% Cu + 0.25% Cr
 - Strength = 504 MPa
 - Used for aircraft structures.

Cast Aluminum Alloys

Aluminum, 99.00% minimum and greater	1xx.x
Aluminum alloys grouped by major alloying elements:	
Copper	2xx.x
Silicon, with added copper and/or magnesium	3xx.x
Silicon	4xx.x
Magnesium	5xx.x
Zinc	7xx.x
Tin	8xx.x
Other element	9xx.x
Unused series	6xx.x

Temper Designation for Aluminum Alloys

- In addition to composition, the properties of aluminum alloys can be modified by heat treatment and mechanical working
- These treatments are expressed in terms of temper designations
 - F – As fabricated
 - O – Annealed
 - H – Strain hardened
 - T – Heat treated to produce a stable temper
 - Natural aging: precipitation treatment at room temperature
 - Artificial aging: precipitation treatment at an elevated temperature
 - For example AA2024-T4 or AA6061-T6

Temper Designations

- H designations
 - H1x – Strain hardened
 - H2x – Strain hardened and partially annealed
 - H3x – Strain hardened followed by a low temperature thermal treatment to improve ductility
 - In the above “x” indicates amount of strain hardening (x=8 means UTS that is achieved by 75% cold work; x=0 means fully annealed; x=4 means UTS half-way between x=0 and x=8)
- T designations
 - T1 – cooled from shaping temperature and naturally aged
 - T2 – cooled from shaping temperature, cold worked and naturally aged
 - T3 – Solution treated, cold worked and naturally aged
 - T4 – Solution treated and naturally aged
 - T5 – Cooled from shaping temperature and artificially aged
 - T6 – Solution treated and artificially aged
 - T7 – Solution treated and overaged – improves resistance to stress corrosion cracking
 - T8 – Solution treated, cold worked and artificially aged

Aluminum Association Number	UNS Number	Composition (wt%) ^a	Condition (Temper Designation)	Mechanical Properties			Typical Applications/ Characteristics
				Tensile Strength [MPa (ksi)]	Yield Strength [MPa (ksi)]	Ductility [%EL in 50 mm (2 in.)]	
Wrought, Nonheat-Treatable Alloys							
1100	A91100	0.12 Cu	Annealed (O)	90 (13)	35 (5)	35–45	Food/chemical han- dling & storage equipment, heat ex- changers, light re- flectors
3003	A93003	0.12 Cu, 1.2 Mn, 0.1 Zn	Annealed (O)	110 (16)	40 (6)	30–40	Cooking utensils, pressure vessels and piping
5052	A95052	2.5 Mg, 0.25 Cr	Strain hardened (H32)	230 (33)	195 (28)	12–18	Aircraft fuel & oil lines, fuel tanks, ap- pliances, rivets, and wire
Wrought, Heat-Treatable Alloys							
2024	A92024	4.4 Cu, 1.5 Mg, 0.6 Mn	Heat treated (T4)	470 (68)	325 (47)	20	Aircraft structures, rivets, truck wheels, screw ma- chine products
6061	A96061	1.0 Mg, 0.6 Si, 0.30 Cu, 0.20 Cr	Heat treated (T4)	240 (35)	145 (21)	22–25	Trucks, canoes, rail- road cars, furni- ture, pipelines
7075	A97075	5.6 Zn, 2.5 Mg, 1.6 Cu, 0.23 Cr	Heat treated (T6)	570 (83)	505 (73)	11	Aircraft structural parts and other highly stressed ap- plications

UNS – A9 used to identify wrought aluminum alloys

Aluminum Association Number	UNS Number	Composition (wt%) ^a	Condition (Temper Designation)	Mechanical Properties			Typical Applications/ Characteristics
				Tensile Strength [MPa (ksi)]	Yield Strength [MPa (ksi)]	Ductility [%EL in 50 mm (2 in.)]	
Cast, Heat-Treatable Alloys							
295.0	A02950	4.5 Cu, 1.1 Si	Heat treated (T4)	221 (32)	110 (16)	8.5	Flywheel and rear- axle housings, bus and aircraft wheels, crankcases
356.0	A03560	7.0 Si, 0.3 Mg	Heat treated (T6)	228 (33)	164 (24)	3.5	Aircraft pump parts, automotive trans- mission cases, wa- ter-cooled cylinder blocks
Aluminum-Lithium Alloys							
2090	—	2.7 Cu, 0.25 Mg, 2.25 Li, 0.12 Zr	Heat treated, cold worked (T83)	455 (66)	455 (66)	5	Aircraft structures and cryogenic tank- age structures
8090	—	1.3 Cu, 0.95 Mg, 2.0 Li, 0.1 Zr	Heat treated, cold worked (T651)	465 (67)	360 (52)	—	Aircraft structures that must be highly damage tolerant

UNS – A0 used to identify cast aluminum alloys

Magnesium Alloys

- Density $\sim 1.74 \text{ g/cm}^3$, less than that of Al (2.7 g/cm^3)
- More expensive than aluminum because
 - HCP structure makes Mg difficult to cold work – hot work only
 - Molten Mg can burn in air – difficult to cast
- Classification:
 - Two letters followed by two numbers
 - A – Aluminum
 - K – Zirconium
 - M – Manganese
 - E – Rare Earth
 - H – Thorium
 - Q – Silver
 - S – Silicon
 - T – Tin
 - Z – Zinc
 - The numbers indicate approximate alloying content
 - Additional letters to indicate variations of the basic alloy
- Temper classification similar to aluminum alloys

ASTM Number	UNS Number	Composition (wt%) ^a	Condition	Mechanical Properties			Typical Applications
				Tensile Strength [MPa (ksi)]	Yield Strength [MPa (ksi)]	Ductility [%EL in 50 mm (2 in.)]	
Wrought Alloys							
AZ31B	M11311	3.0 Al, 1.0 Zn, 0.2 Mn	As extruded	262 (38)	200 (29)	15	Structures and tubing, cathodic protection
HK31A	M13310	3.0 Th, 0.6 Zr	Strain hardened, partially annealed	255 (37)	200 (29)	9	High strength to 315°C (600°F)
ZK60A	M16600	5.5 Zn, 0.45 Zr	Artificially aged	350 (51)	285 (41)	11	Forgings of maximum strength for aircraft
Cast Alloys							
AZ91D	M11916	9.0 Al, 0.15 Mn, 0.7 Zn	As cast	230 (33)	150 (22)	3	Die-cast parts for automobiles, luggage, and electronic devices
AM60A	M10600	6.0 Al, 0.13 Mn	As cast	220 (32)	130 (19)	6	Automotive wheels
AS41A	M10410	4.3 Al, 1.0 Si, 0.35 Mn	As cast	210 (31)	140 (20)	6	Die castings requiring good creep resistance

*The balance of the composition is magnesium.

Source: Adapted from *ASM Handbook*, Vol. 2, *Properties and Selection: Nonferrous Alloys and Special-Purpose Materials*, 1990. Reprinted by permission of ASM International, Materials Park, OH.

UNS – Letter M indicates magnesium alloys

Titanium Alloys

- Titanium is the 4th most common metal on the earth's crust.
 - Chemically very reactive and is difficult to extract
 - Like Cr and Al, it forms a protective oxide layer, making it corrosion resistant
 - Density $\sim 4.5 \text{ g/cm}^3$ – lower density than Fe or Ni, higher use temperature than Al
 - Exhibits polymorphism:
 - At low temperatures: Alpha α – hcp
 - At high temperatures: Beta β – bcc
- Alloying elements are either
 - Alpha stabilizers – Al, O make the alpha phase stable at higher temperatures
 - Beta stabilizers – V, Mo, Fe and Cr cause a eutectoid reaction in the alloys and make the beta phase to be stable at lower temperatures, even down to RT
- Alloys classified as α , β or $\alpha+\beta$ depending on the composition
- New alloys are still being developed, and UNS designations have not been standardized for all alloys
- Properties depend upon composition and thermomechanical processing that can change the microstructure of the alloys
- Processing of titanium alloys is very difficult because of the structure
- Expensive aerospace alloy that is now seeing more commercial applications

Alloy Type	Common Name (UNS Number)	Composition (wt%)	Condition	Average Mechanical Properties			Typical Applications
				Tensile Strength [MPa (ksi)]	Yield Strength [MPa (ksi)]	Ductility [%EL in 50 mm (2 in.)]	
Commercially pure	Unalloyed (R50500)	99.1 Ti	Annealed	484 (70)	414 (60)	25	Jet engine shrouds, cases and airframe skins, corrosion-resistant equipment for marine and chemical processing industries
α	Ti-5Al-2.5Sn (R54520)	5 Al, 2.5 Sn, balance Ti	Annealed	826 (120)	784 (114)	16	Gas turbine engine casings and rings; chemical processing equipment re- quiring strength to temperatures of 480°C (900°F)
Near α	Ti-8Al-1Mo- 1V (R54810)	8 Al, 1 Mo, 1 V, balance Ti	Annealed (duplex)	950 (138)	890 (129)	15	Forgings for jet engine components (compressor disks, plates, and hubs)
α - β	Ti-6Al-4V (R56400)	6 Al, 4 V, balance Ti	Annealed	947 (137)	877 (127)	14	High-strength prosthetic implants, chemical-processing equipment, air- frame structural components
α - β	Ti-6Al-6V-2Sn (R56620)	6 Al, 2 Sn, 6 V, 0.75 Cu, balance Ti	Annealed	1050 (153)	985 (143)	14	Rocket engine case airframe applica- tions and high-strength airframe structures
β	Ti-10V-2Fe-3Al	10 V, 2 Fe, 3 Al, balance Ti	Solution + aging	1223 (178)	1150 (167)	10	Best combination of high strength and toughness of any commercial tita- nium alloy; used for applications re- quiring uniformity of tensile proper- ties at surface and center locations; high-strength airframe components

Source: Adapted from *ASM Handbook, Vol. 2, Properties and Selection: Nonferrous Alloys and Special-Purpose Materials*, 1990.
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UNS – Letter R indicates refractory metal (high melting point)
R5xxxx – Titanium alloys

Materials Selection

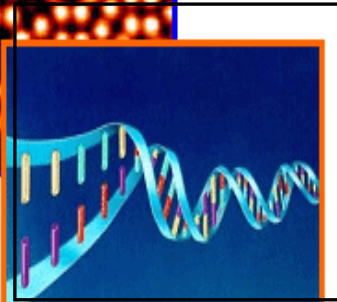
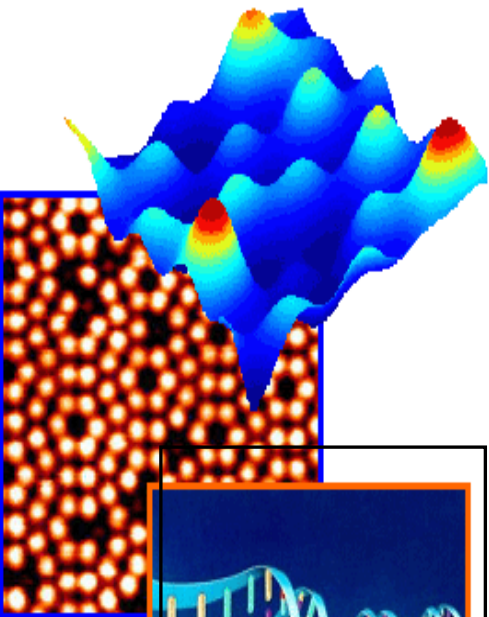
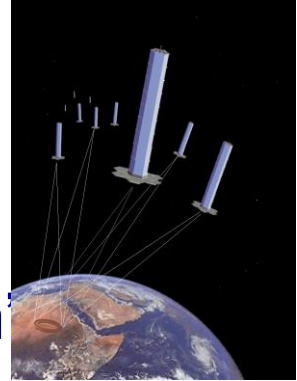
- Mechanical properties
 - Stiffness, strength, ductility, fatigue, creep
- Manufacturability
 - Machining, Mechanical working, Casting, Welding
- Physical properties
 - Density, Melting point, Thermal conductivity
- Cost
 - Availability, ease of processing

Smart Materials for Sensing & Actuation

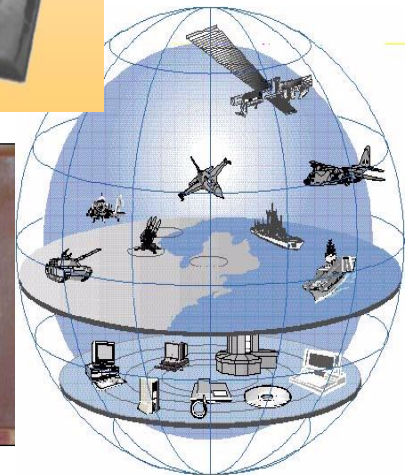
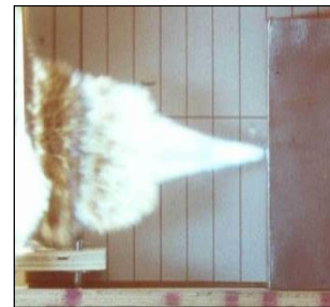
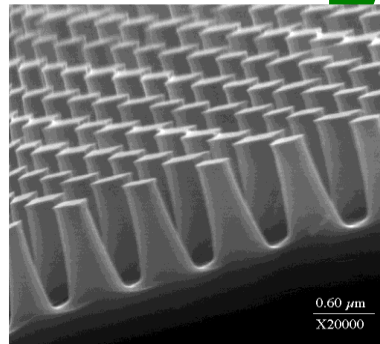
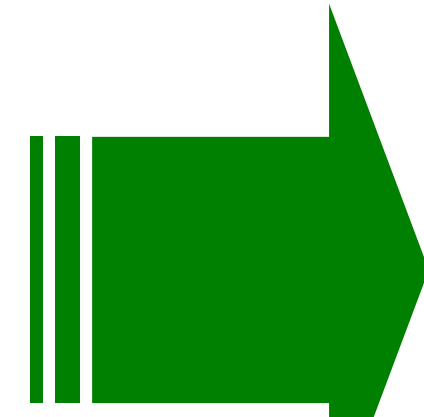
Output	Current/ Charge	Magne- tization	Strain	Temperature	Light
Input					
Electric Field	Conductivity Permittivity	Electromagnetic Effect	Reverse Piezoelectricity	Ohmic Resistance	Electro-Optic effect
Magnetic Field	Eddy Current Effect	Permeability	Joule Effect Magnetostriction	Magneto caloric Effect	Magnet o-Optic effect
Stress	Direct Piezo-Electric Effect	Villary Effect	Elastic Modulus	Thermo-Mechanical Effect	Photo-elastic Effect
Heat	Pyro-electric Effect	Thermo-Magne-tization	Thermal Expansion Phase Transition	Specific Heat	Thermo-Lumin-escence
Light	Photo-Voltaic Effect	Photo-Magne-tization	Photo-striction	Photo-Thermal effect	Refract-ive index

Nano Science & Technology

“Technology based on the ability to manipulate and control elements with dimensions < 100 nm



Focused R&D



Revolutionary System Capabilities

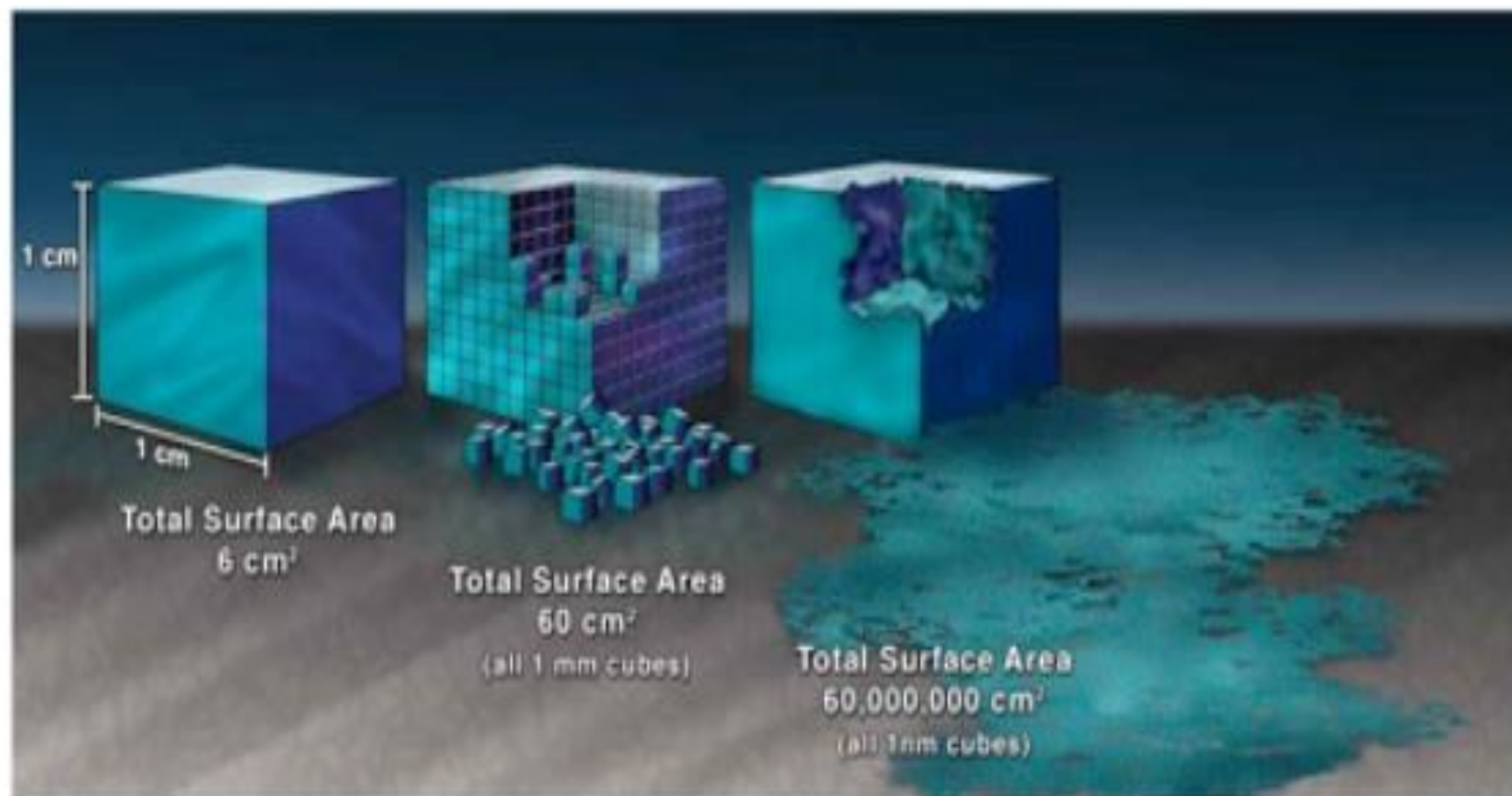
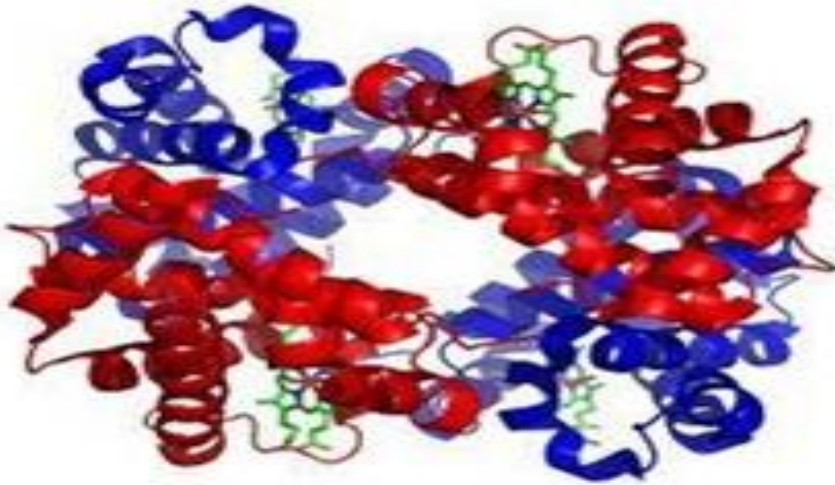
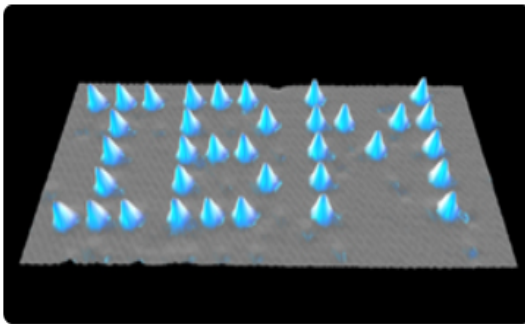


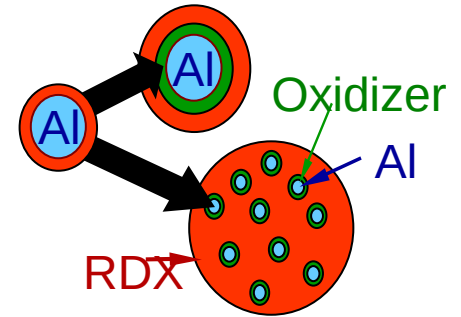
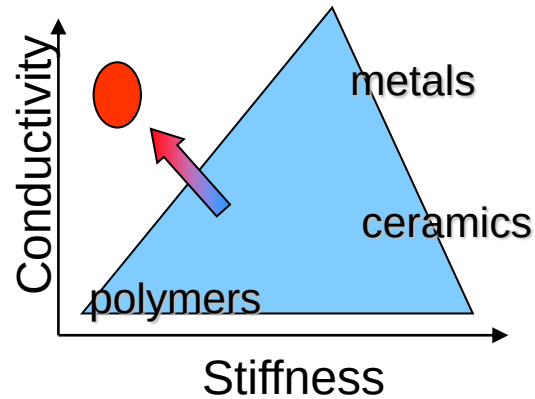
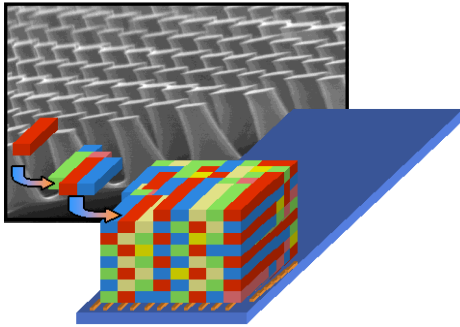
Illustration demonstrating the effect of the increased surface area provided by nanostructured materials.



Computer simulation of hemoglobin, a naturally occurring nanoscale protein that is found in blood. (Image: Centers for Disease Control)



1989: Don Eigler and Erhard Schweizer at IBM's Almaden Research Center **manipulated 35 individual xenon atoms to spell out the IBM logo.** This demonstration of the ability to precisely manipulate atoms ushered in the applied use of nanotechnology. (Image at left.)



Nano Structured
Devices

Nano Engineered
Materials

Nano
Energetics

Foundation & Tools:
Modeling - Nano Assembly & Fabrication - Characterization

Nano Engineered Materials

Potential Return on Investment

Investments in

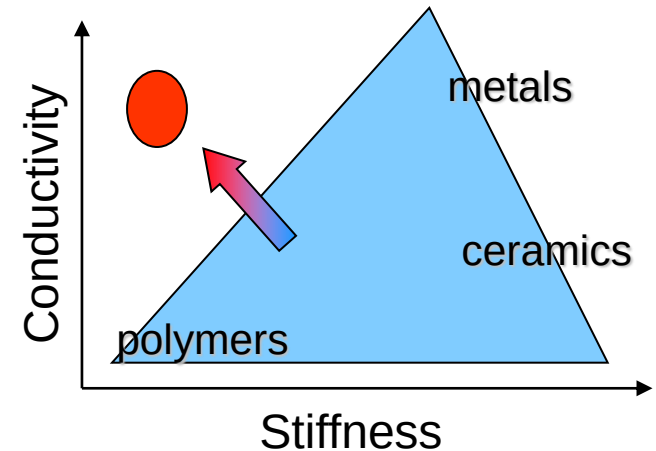
Nanoengineered composites

High-energy-density materials

Adaptive & self-healing materials

Nanostructured electronic/OE materials

Quantum mechanical & molecular
dynamics modeling



Offer Potential For

10-100x stronger materials - lightweight aerospace vehicles

2x increase in temperature, 100x lifetime materials for turbine engines

Ability to simultaneously optimize multiple desired materials characteristics, e.g.
electrical and structural characteristics

Adaptive materials - dynamic stealth, improved survivability of aerospace assets

Nano Energetics

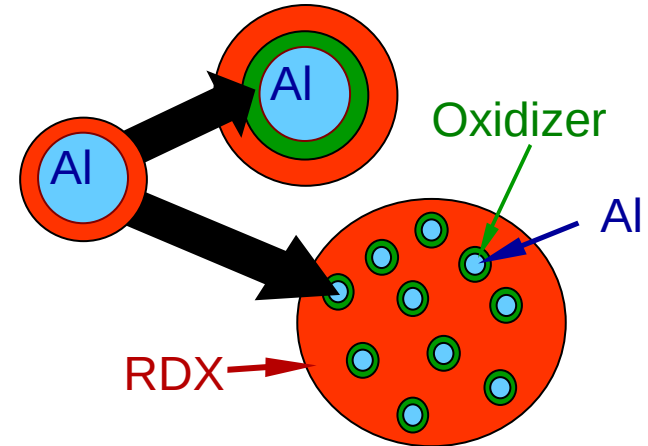
Potential Return on Investment

Investments in

- Nanostructured explosives and fuel additives
- Nano controlled catalytics
- Nano engineered photovoltaics

Offer Potential For

- 10-100x improved power density/energy release rate/ package volume and increased safety – miniature smart munitions
- Improved munitions blast control – better coupling of energy to target, lower collateral damage
- Stable, 5-10x more efficient propellants – safer, high-thrust-to-weight-ratio propulsion systems, increased weapon loadout, affordable access to space
- Improved power generation, conditioning & control for aerospace vehicles – enhanced global reach

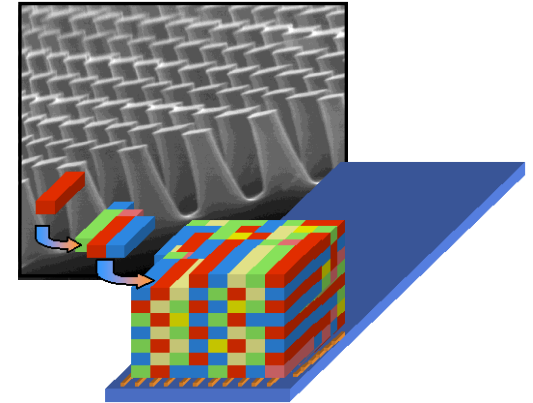


Nano Structured Devices

Potential Return on Investment

Investments in

- Nano processing/comm devices, circuits, ICs
- Nano actuators, controllers, sensors, photonics & integrated NEMS
- Revolutionary computing architectures – quantum, molecular, optical
- Quantum communications & cryptography

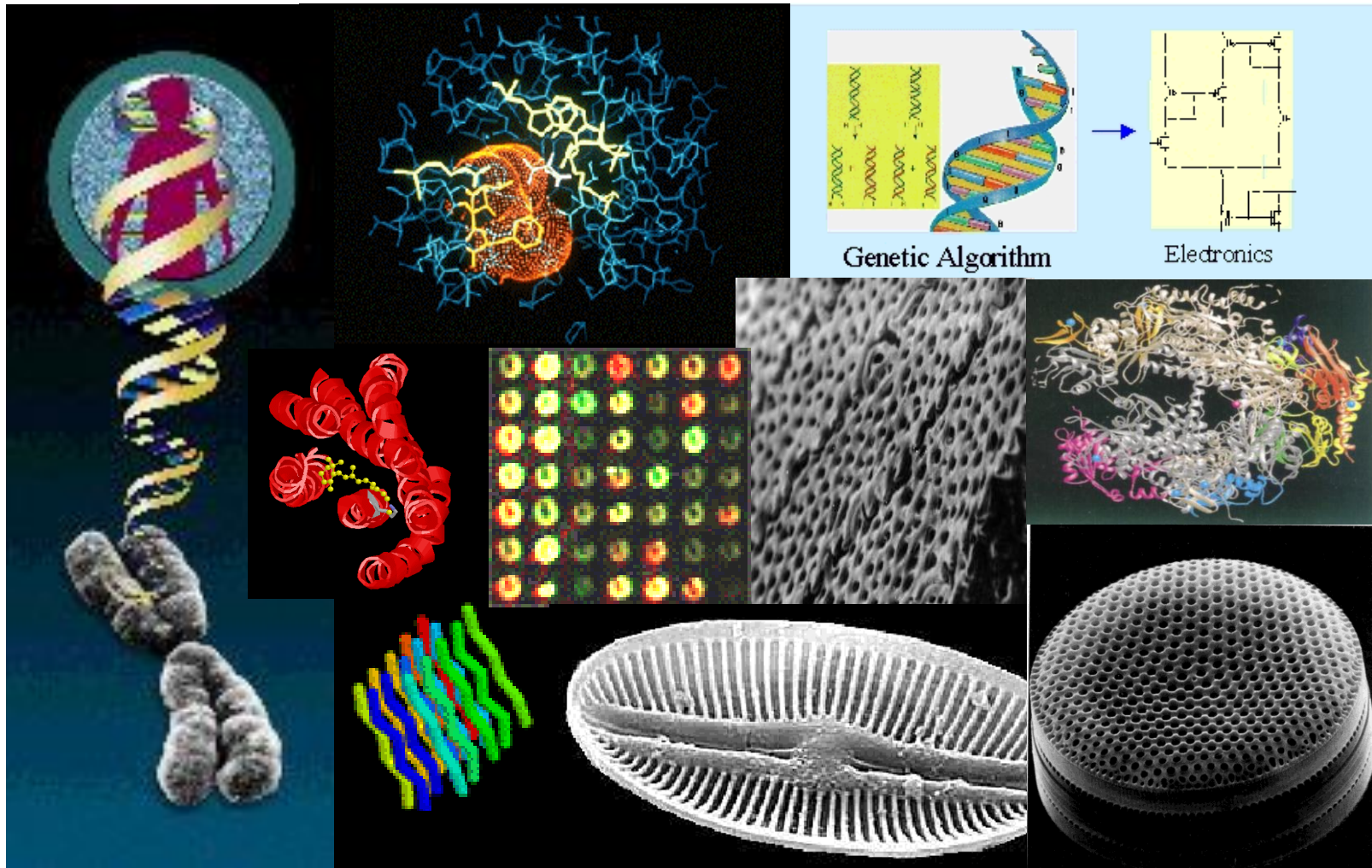


Offer Potential For

- 10-1000x higher speed, lower power higher density electronics – on-board processing for target tracking/ID and autonomy
- Exponential speed up for hard problem solution, e.g. scheduling, multi-asset control
- 10-1000x smaller sensors, actuators & NEMS – new surveillance capabilities, e.g. microsat constellations, UAVs, UCAVs and large, space-based membrane apertures
- High-bandwidth, secure communications - information dominance

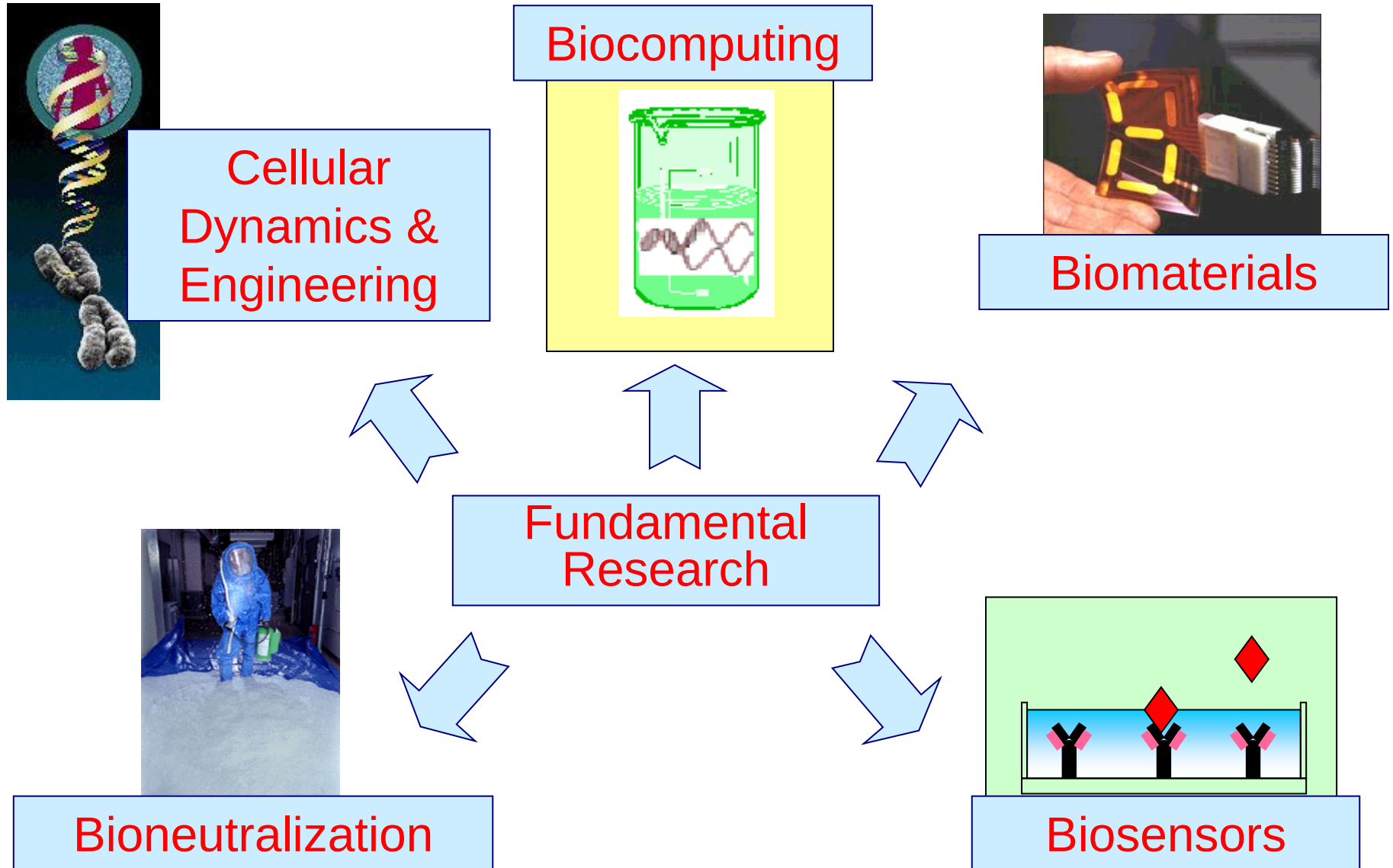
Biotechnology

“The ways and means of exploiting biological systems at the molecular level, to provide new products and capabilities.”





Biotechnology Focus Areas



Aerospace Biotechnology

Proposed Investment Areas

Biomaterials

Bottoms-up fabrication: Biocatalysis and synthesis by microorganisms

Smart skins: Dynamic, biomimetic, environmentally-adaptable coatings

Bio-inspired materials: Self-powering, self-assembling materials & structures

Bio-energetics: Charge separation, charge pumping, and energy transduction

Fundamental Research

Toxigenomics

Biocomputing

Biologically based concepts for data storage and processing

Bionutralization

Decontamination strategies for chemical and biological agents

Methodologies to validate bionutralization

Biosensors

Non-cryo IR sensors incorporating biomolecular recognition and biomimetic strategies

Simultaneous, rapid multiple bio-threat point detectors

Cellular Dynamics & Engineering

Control systems for intelligent affordable, robust autonomous systems

Aerospace Biotechnology

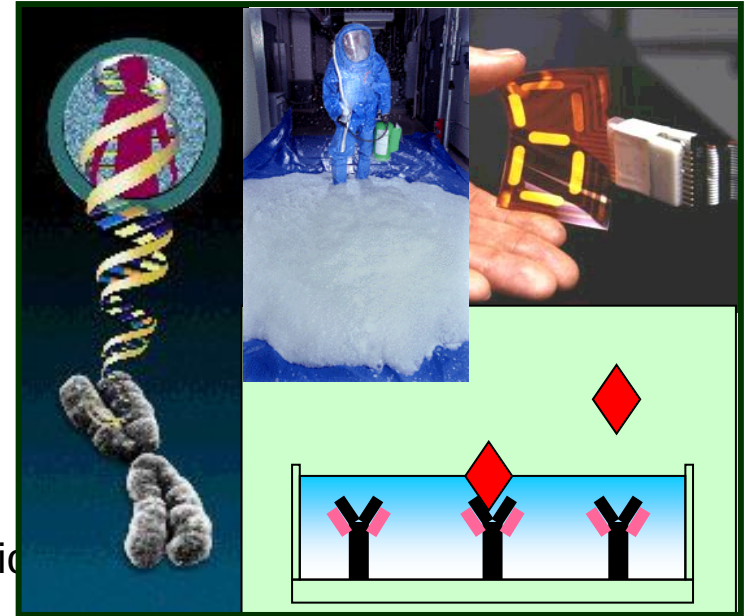
Potential Investment Payoff

Investments in

- Understanding biological sensing mechanisms
- Distributed sensing and actuation systems
- Neural network architectures
- DNA information storage and processing
- Biomimetic control architectures
- Self-healing and adaptation mechanisms

Offer Potential For

- 10-1000x more sensitive detection and signal extraction
- 1000x higher information storage density
- 1000x lower power and higher speed computational systems
- Radically improved decision making
- Greatly simplified control architectures for complex and distributed systems
- Adaptable, self protecting, self diagnosing and self-healing space systems
- Self-assembled complex systems



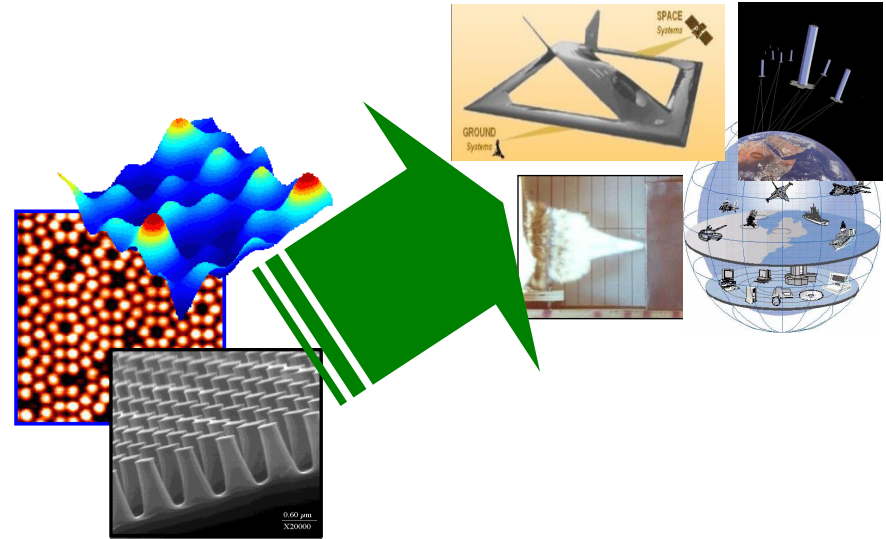
Nano and Bio Technology

Nano Science & Technology

Nano engineered materials

Nano energetics

Nano structured devices



Aerospace Biotechnology

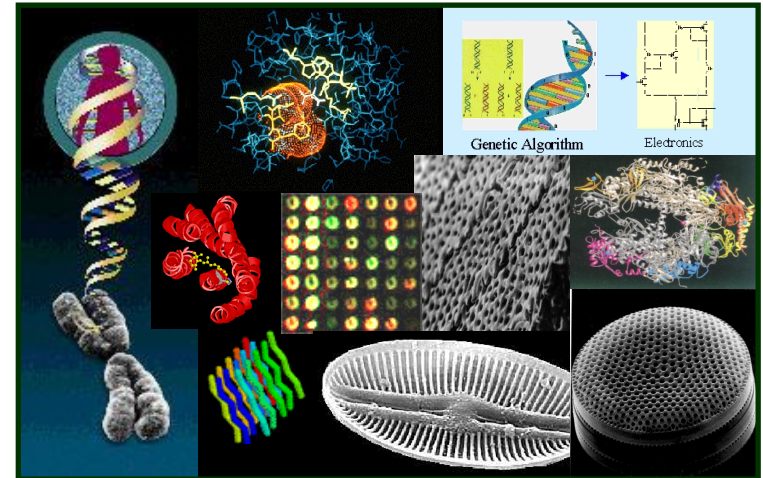
Cellular dynamics & engineering

Bio materials

Bio sensors

Bio neutralization

Bio computing



Bulk Material

Top Down Approach

- ❖ Mechanical Milling
- ❖ Etching
- ❖ Laser Ablation
- ❖ Sputtering
- ❖ Electro-explosion

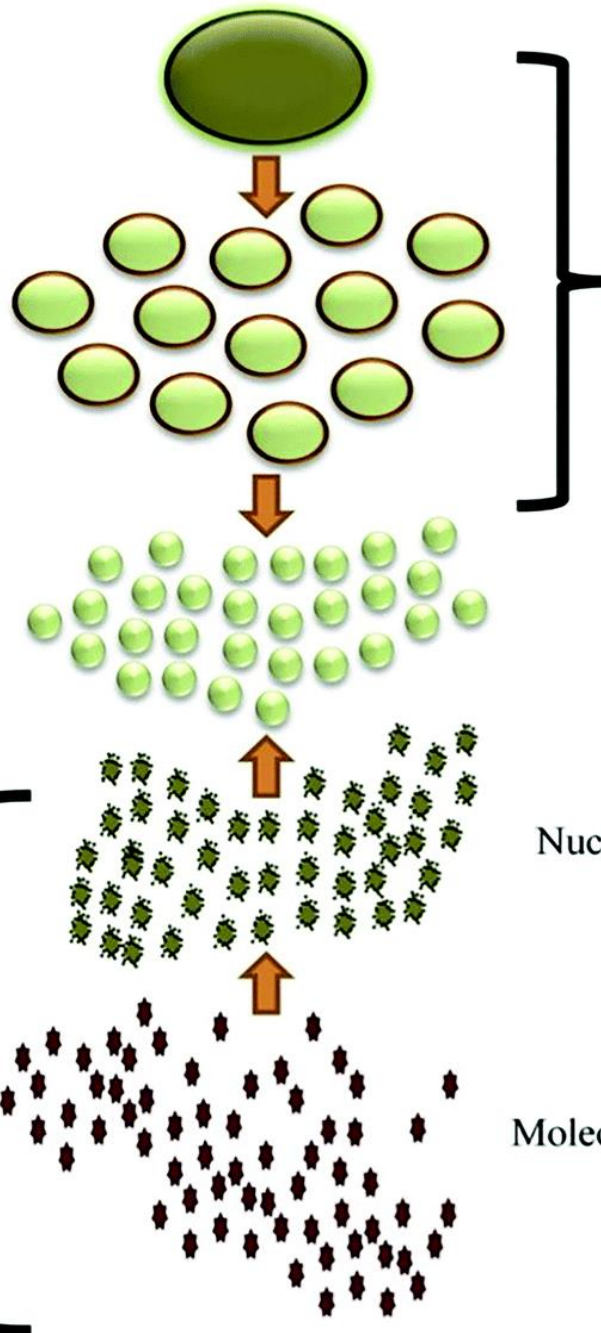
Nanoparticles

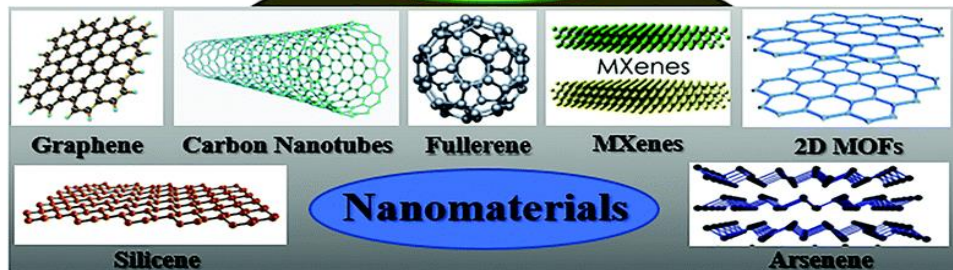
Bottom Up Approach

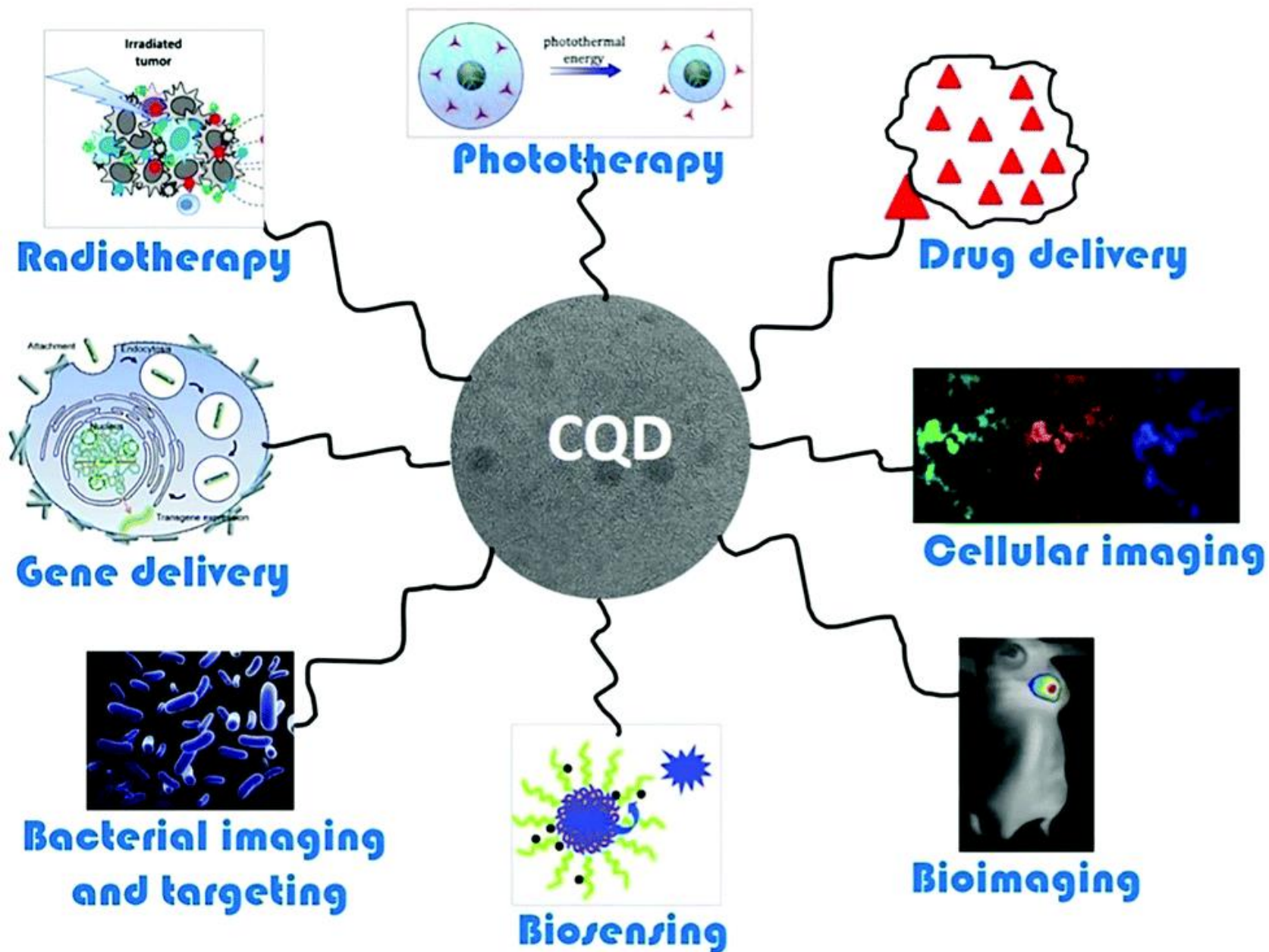
- ❖ Supercritical Fluid Synthesis
- ❖ Spinning
- ❖ Sol-gel Process
- ❖ Laser Pyrolysis
- ❖ Chemical Vapour Deposition
- ❖ Molecular Condensation
- ❖ Chemical Reduction
- ❖ Green Synthesis

Nuclei and its growth

Molecular/Atomic Level

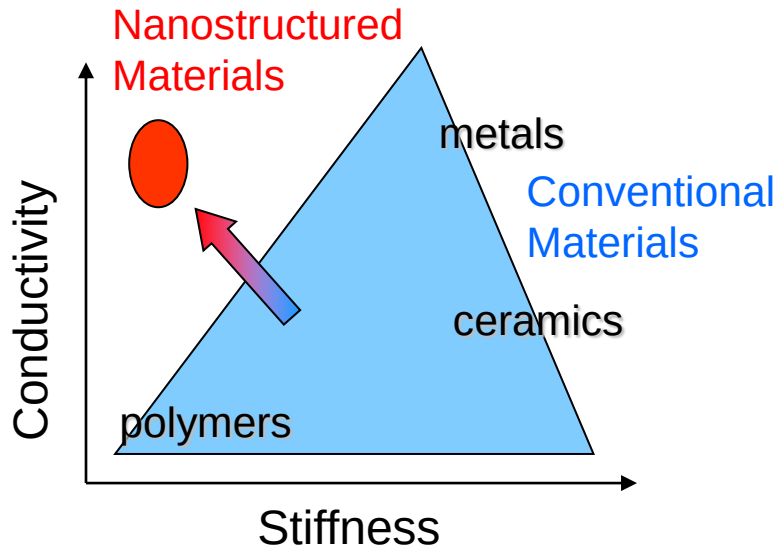






Nano Engineered Materials

Example: Conductive Elastomers



Materials by Design

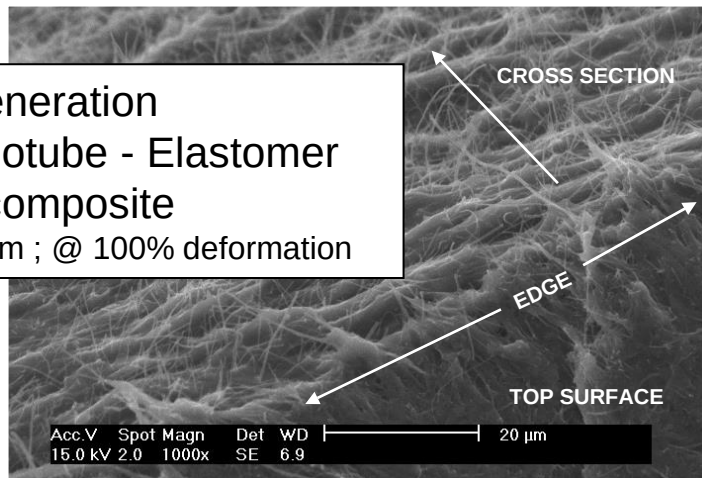
Control of material structure at the nanoscale avoids classic performance-processing trade-offs, enabling the design and manufacture of revolutionary new materials with unique *combinations of properties*

- Independently tailorable structural, thermal, electrical, optical and/or magnetic properties
- Dynamic properties, e.g. self-diagnosing & self-healing materials

Payoff to the Warfighter

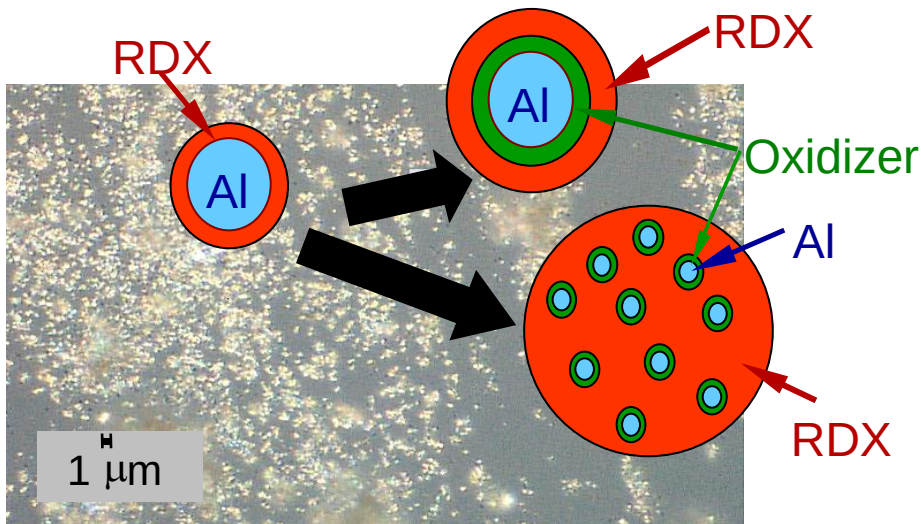
- Improved LO maintainability & dynamic stealth
- Broadband, lightweight EM shielding
- Lower weight, higher temperature engine materials for greater thrust and longer range

1st Generation
C-Nanotube - Elastomer
Nanocomposite
1-10 S/cm ; @ 100% deformation



Nano Energetics

Example : Nano Structured Explosives

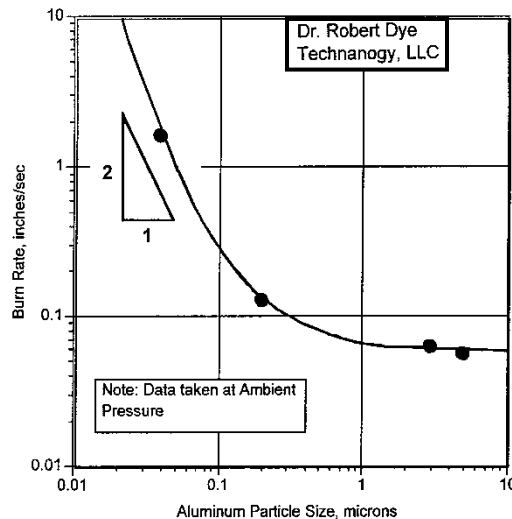


Nanoscale Aluminum, Coated by RDX

Nano fuel particles coated by oxidizer

- 100x power – increase in energy release rate
- 2x total energy – greater surface and internal volume free energy available
- 10x efficiency – near 100% complete reactions
- 10x safer – lower sensitivity to mechanical initiation
- More compact - no binder

Burn Rate vs.
Particle Size



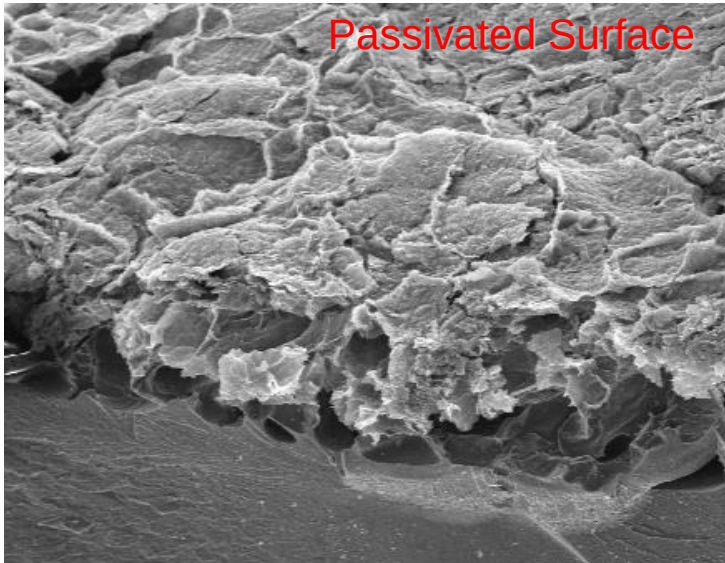
Payoff to the Warfighter

- Smaller, safer munitions
- More kills per sortie
- Decreased logistics tail
- Enables small, weaponized UAVs

Nano Engineered Materials Example

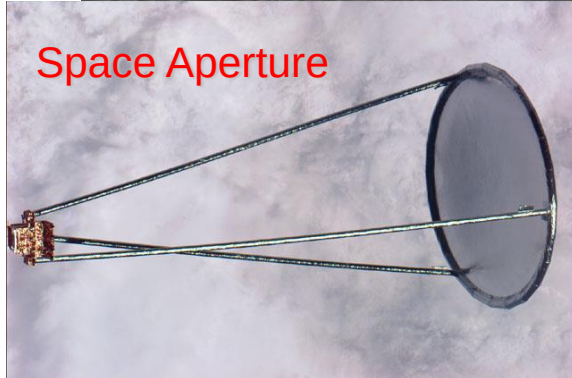
Pay off in Aerospace Structures

Passivated Surface



Space Aperture

200 μm



Solid Rocket Motor



Technology Description

Control distribution and arrangement of nanoparticles in polymers

Generate desirable new properties while maintaining processability & low cost

- 2x higher temperature operation
- 100x lifetime - self-passivating surfaces
- Insensitivity to space environment

Benefits to the Warfighter

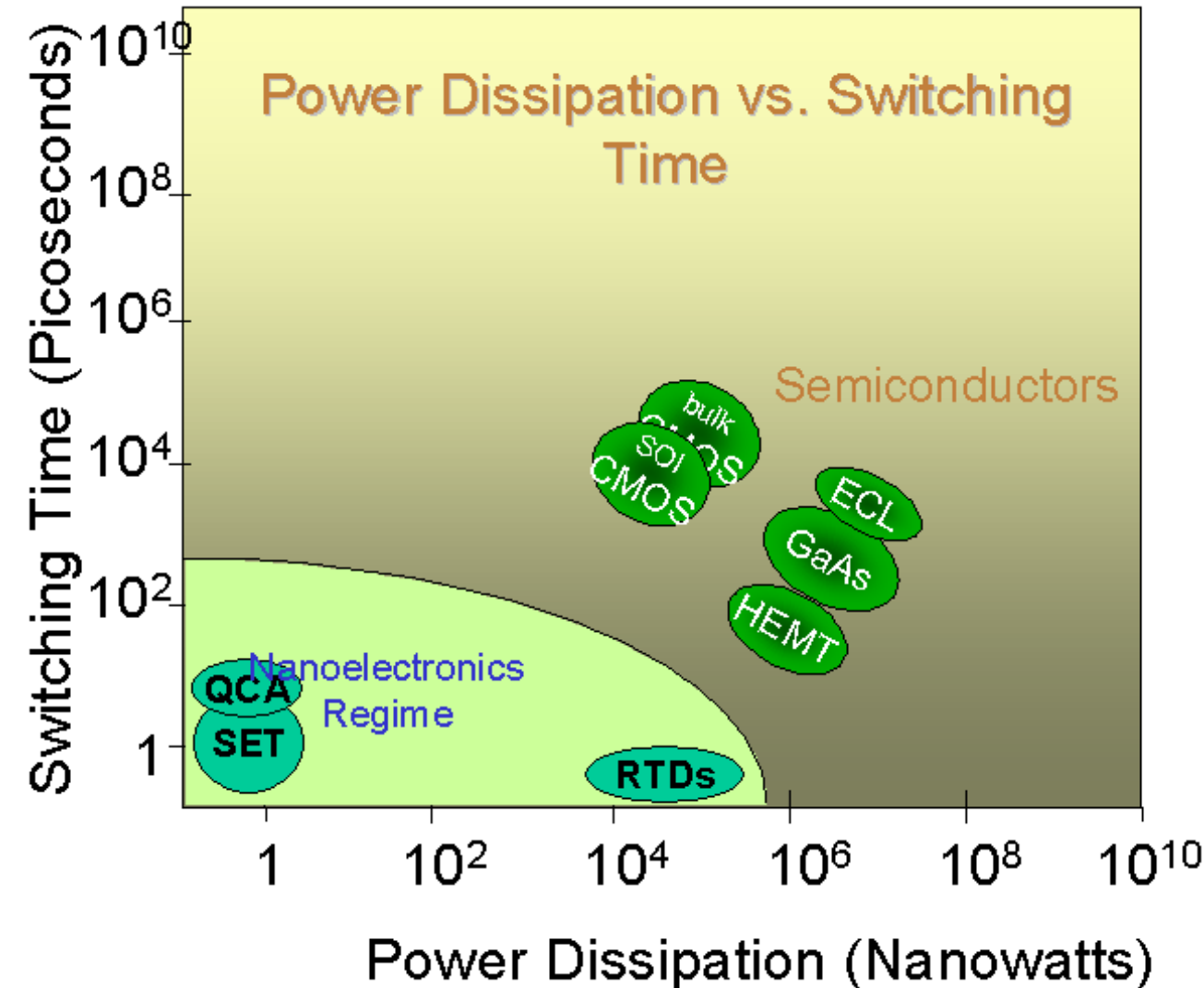
Provides 10-100x greater service life for rocket and space environments

Superior insulation material for solid rocket & high-temperature turbines

Low-mass, low-cost structural material for large space apertures

Nano Structured Devices

Example : High-Speed Devices



Processing Power

Nano devices offer the potential for orders of magnitude increase in speed and simultaneous reduction in power

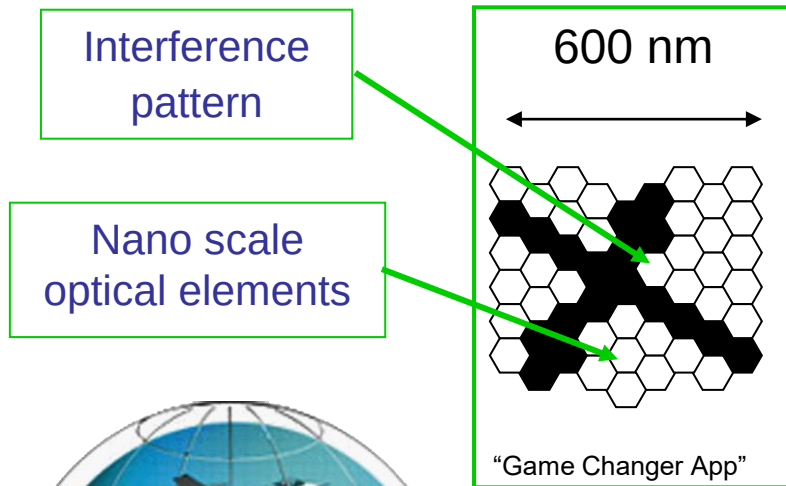
Payoff to the Warfighter

- Orders of magnitude increase in on-board processing capability
- Real time information extraction from complex sensor systems to aid decision makers

Nano Structured Devices Example

Payoff in New Display Capability

Nano Scale Holographic Grating



Digitizing Holographic Gratings

Exploit coherent electromagnetic scattering from nano structures to generate an immersive, 3D holographic display

Benefits to the Warfighter

True 3D representation of the battlespace for enhanced tactical and strategic battlespace awareness

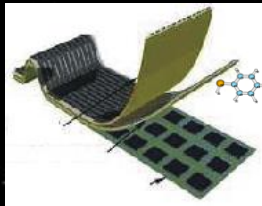
More realistic heads-up displays for pilot situational awareness

Superior combat training simulations

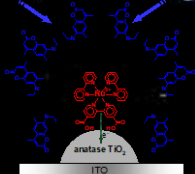
System Example:

Generation after Next Surveillance Satellite

Nano-engineered electronic materials
(self protecting)

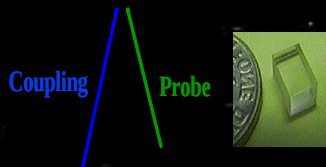


Molecular photovoltaics
(2x SOA efficiency)

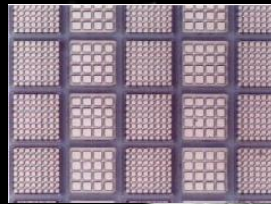


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Quantum-computing
(on platform analysis)

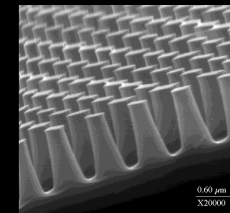


Broadband, reconfigurable sensors
(rapid, unambiguous target ID)

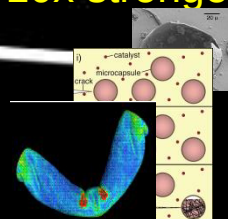


Large membrane, nano-controlled apertures
(10x lighter)

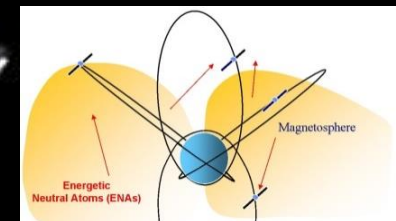
Reconfigurable EO properties
(dynamic stealth)



Self-healing materials
(10x lighter, 10x stronger)



Nano sensors
(ubiquitous hazard alert)



Nano/Bio/Info Convergence: *The 21st Century Revolution*

